AI BUILDER CUP 2026, JAPAC

Training Failure Triage

11 / 14

Evidence for deciding whether a training failure comes from software or GPU hardware, with risk controls for the proposed action.

Detection | Evidence collection | Diagnosis and plan | Action and verification

11 / 14 is the joint classification and RMA rule score on 14 constructed cases

gpu-waste-auditor-2026.web.app

Sustainability & Social Impact

Submission overview

Roadmap

Training failure diagnosis and controlled actions

The deck covers the Xid 94 contrast, the browser application, and a reproducible rule benchmark. Historical patrol measurements describe individual runs.

Six simulated nodes. Human approval for medium and high risk actions.

01 The problem
Hardware escalation depends on error frequency and ECC trends
03 – 05

02 Application workflow
Cluster inspection and action approval with state verification
06 – 10

03 Evidence and limits
Rule baseline of 11/14 and the unsupported artifact tier
11 – 15

CHAPTER 01

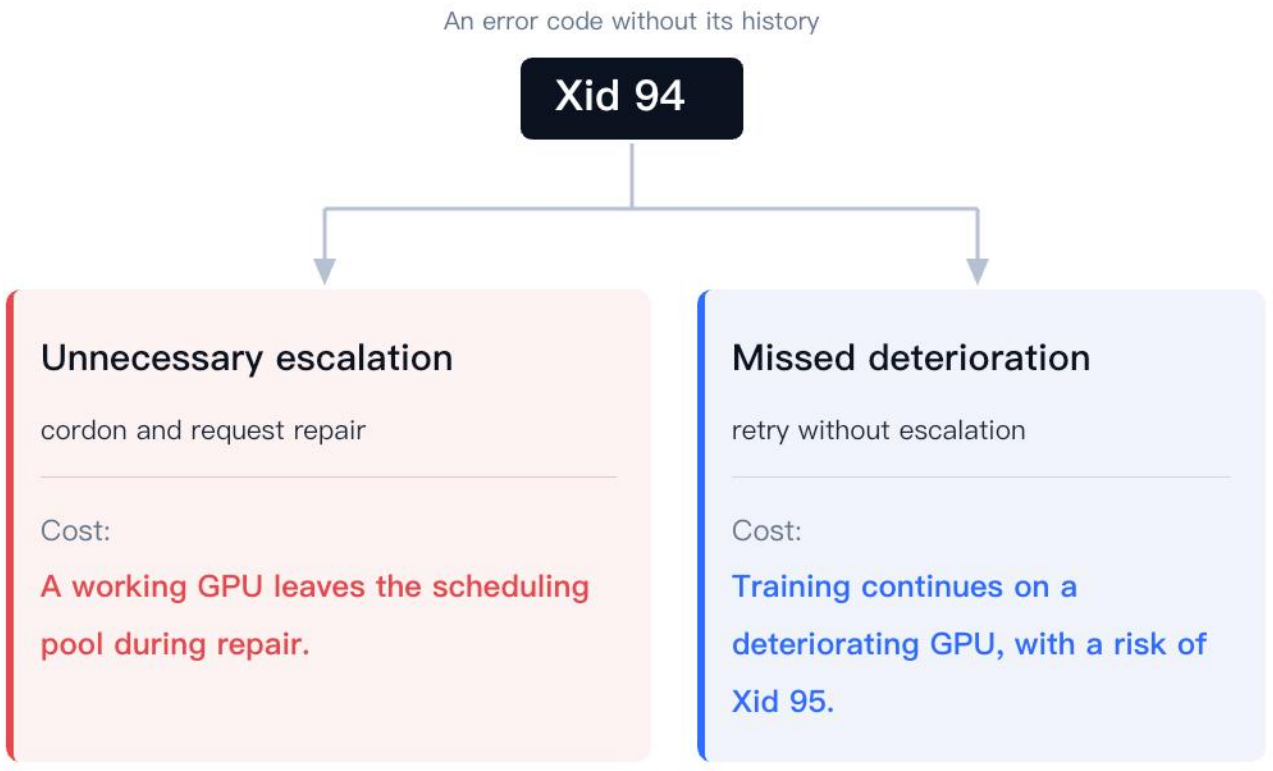
The problem

Hardware escalation after a training failure

94

Unnecessary repair removes a working GPU from the pool. Missed hardware deterioration can cause repeated training failures. Error frequency and ECC history help distinguish these cases.

Xid 94 escalation errors



Evidence needed for escalation

- Xid 94 reports a contained memory error. The code alone does not establish whether hardware repair is required.
- The two nodes differ in their 30-day Xid count and the direction of correctable ECC.
- The rule baseline uses weighted signature matches. These cases also require an assessment of severity over time.

The 30-day history changes the repair assessment

Xid 94 frequency and ECC trends

lyg2103 vs lyg2112, 30-day window

1

lyg2103

Xid 94 in the last 30 days

Correctable ECC 1, flat trend

hardware_watch, no RMA

53

lyg2112

Xid 94 in the last 30 days

Correctable ECC 118, rising trend

hardware, RMA recommended

lyg2103	H800_80G · healthy Xid 94 (Contained memory error) ×1, GPU 5; latest 2026-10-14T01:22:00Z Xid 94 count over 30 days: 1 Current correctable ECC count: 1	Jobs: job-7741(htli)

The source list has two intervening rows, lyg2107 and lyg2108, omitted here

lyg2112	H800_80G · healthy Xid 94 (Contained memory error) ×41, GPU 3; latest 2026-10-14T03:58:00Z Xid 94 (Contained memory error) ×12, GPU 3; latest 2026-10-13T21:10:00Z Xid 94 count over 30 days: 53 Correctable ECC trend: rising Current correctable ECC count: 118	Jobs: job-7903(chenl)

Both nodes report Xid 94. lyg2112 has more frequent events and rising correctable ECC.

English UI capture from the local seed replay, using the same 30-day window

CHAPTER 02

Application workflow

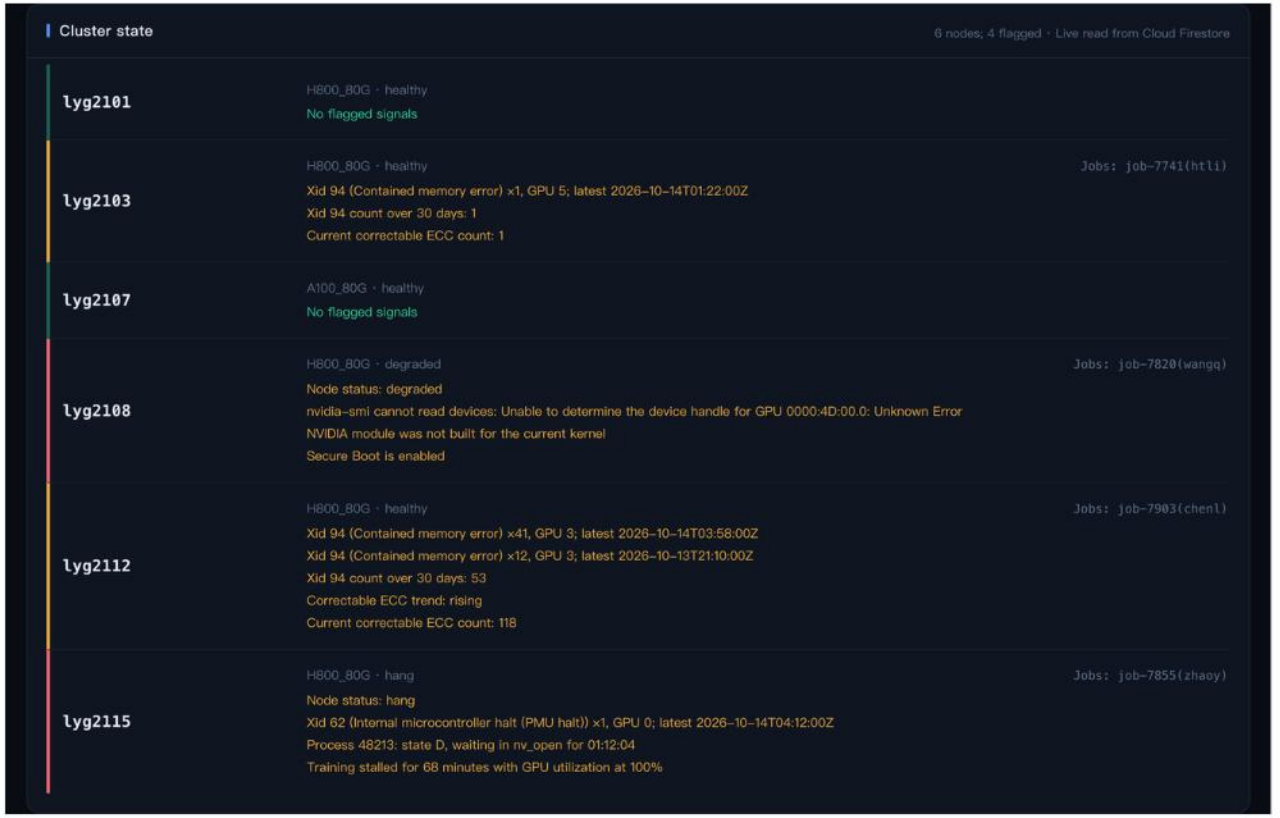
A patrol starts with a human click

17

The recorded demo patrol made 17 tool calls. It inspected six nodes and consulted the knowledge base before taking low risk actions. Tickets remain drafts and notifications remain queued. Each model attempt allows up to 12 turns, and fallback carries completed calls into the next attempt.

Cluster inspection

English UI, local seed replay



The English application renders six seed nodes with four anomaly signals. This capture uses local Firestore response fixtures and makes no external writes.

Initial simulated state

6 nodes in the constructed demo cluster

4 nodes with anomaly signals in the seed

0 log lines pasted for a patrol

The deployed application reads simulated node documents from Cloud Firestore. A human starts each patrol. Scheduled operation would require a scheduler integration.

Local English capture preserves the initial clean state

Action verification

Operation log, local shift replay



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verified cordons in the earlier recorded run, replayed here

lyg2112.cordoned	false	to	**true**
lyg2115.cordoned	false	to	**true**

The execution path checks the returned node state before recording verification.

An unchanged value fails verification. The deployed path updates the simulated Firestore cluster. This English capture replays the two cordon events locally and does not run a new patrol.

Both cordons use the low risk execution path

Action risk and approval

LOW	6	Reversible effects limited to a task or scheduling state. eligible for automatic execution
MEDIUM	5	Affects other jobs on the same node, or needs a short interruption. per-item approval + impact acknowledgement
HIGH	3	May remove hardware capacity or affect multiple nodes. approval, change ticket and second confirmation

The catalog also marks raise_nccl_timeout as discouraged. It hides a timeout symptom without resolving its cause and is excluded from model candidates.

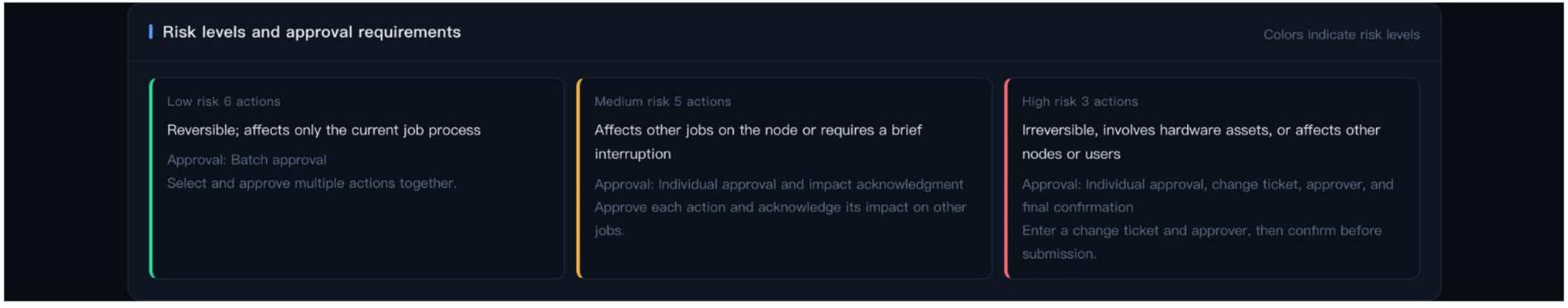
14 actions, 6 low / 5 medium / 3 high

Execution allow-list

```
if (act.risk !== 'low') {
  return { refused: true }
}
```

The model selects an action ID. The action library supplies its risk level.

The tool refuses automatic execution above low risk.



Risk levels and approval requirements from the English application

Triage tool trace

case-xid94-rising, local replay

hardware

RMA advised

0.95

confidence in replay fixture

4 recorded tool calls

expected: hardware, RMA

English replay fixture rendered by the application tool-trace component

```
lookup_xid          {"code":94}
lookup_xid          {"code":95}
inspect_node        {"node":"lyg2103"}
propose_fix         {"action_id":"file_rma","title":"GPU memory deterioration","detail":"Frequent Xid 94 with rising ECC supports an RMA rec
                    ommendation.","classification":"hardware","confidence":0.95,"should_rma":true,"evidence_refs":["T1","T2","T3"]}
Reference map       T1 = lookup_xid T2 = lookup_xid T3 = inspect_node
```

Tool evidence for the recommendation

lookup_xid reads the Xid knowledge base. inspect_node supplies node telemetry. The prompt requires tool evidence for numerical claims.

References T1, T2 and T3

The app registers references when tools execute. It drops unknown reference IDs from proposed actions and reports the dropped IDs.

Local replay demonstrates English rendering and makes no model request

CHAPTER 03

Evidence and limits

A rule benchmark on 14 constructed cases

F8 requires both the classification and RMA decision to match the expected result. The rule baseline scores 11/14. The constructed case set does not establish production accuracy, and this submission reports no aggregate AI score.

11 / 14

rule baseline on the 14–case library

Rule benchmark results

Offline baseline, no model calls

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78.6% joint F8, class and RMA

Three rule-baseline misses

case-xid48-dbe

hardware evidence loses to application signature votes

case-xid94-rising

frequent Xid 94 with rising ECC remains hardware_watch

case-gpu-hang-deadlock

a secondary hang taken for the root cause

Classification alone: 11/14. RMA alone: 12/14. The baseline votes over signature hits and maps hardware to an RMA recommendation.

Per-case comparison — Rule baseline complete (local browser calculation)

CASE	EXPECTED CLASS	EXPECTED REPAIR	RULE DECISION	RULE RESULT	AI DECISION	AI RESULT
case-oom-batch	config	No	config · No repair	Pass	Not run	–
case-oom-fragmentation	config	No	config · No repair	Pass	Not run	–
case-nccl-nvlink	network	No	network · No repair	Pass	Not run	–
case-illegal-access	application	No	application · No repair	Pass	Not run	–
case-driver-old	environment	No	environment · No repair	Pass	Not run	–
case-xid48-dbe	hardware	Yes	application · No repair	Fail	Not run	–
case-xid94-single	hardware_watch	No	hardware_watch · No repair	Pass	Not run	–
case-xid94-rising	hardware	Yes	hardware_watch · No repair	Fail	Not run	–
case-xid79-bus	hardware	Yes	hardware · Repair	Pass	Not run	–
case-gsp-119	environment	No	environment · No repair	Pass	Not run	–
case-driver-nvml-mismatch	environment	No	environment · No repair	Pass	Not run	–
case-smi-driver-dead	hardware	Yes	hardware · Repair	Pass	Not run	–
case-gpu-hang-xid62	environment	No	environment · No repair	Pass	Not run	–
case-gpu-hang-deadlock	application	No	network · No repair	Fail	Not run	–

All 14 constructed cases, scored locally. The AI column shows Not run for this capture.

scripts/verify_triage.mjs and the browser use the same scoring path

Application architecture

01
Knowledge base
24 Xid entries, 27 failure signatures and 6 diagnostic principles

02
Deterministic pre-analysis
Local compatibility checks and memory accounting produce structured facts

03
Tools
10 registered tools, with 6 available for cases and all 10 for patrol

04
Model loop
Gemini allows up to 12 turns per attempt. Fallback retains completed calls.

05
Execution
Automatic actions require low risk. The app checks state and records the result.

Prompt and code responsibilities

The prompt requires tool evidence for numerical claims. The action library supplies risk levels for the action IDs the model selects.

Code limits available tools and rejects automatic execution above low risk. Citation checks remove unknown references from proposed actions.

Medium and high risk actions require operator approval

The app selects each mode's tool list before the model loop starts

Evidence support and REA

3 declared tiers, 2 implemented

1	snapshot	**Case evidence snapshot** Constructed telemetry stored with a case for repeatable diagnosis and benchmark scoring.	SHIPPED
2	live	**Live read** Current node state from the simulated Firestore cluster, including action results.	SHIPPED
3	artifact	**Local artifact inspection** Inspection of local files such as binaries or packet captures requires a separate deployment.	NO PRODUCER

Browser deployment boundary

The submitted browser build has no local artifact producer. The artifact tier has implemented set to false. Requests for it raise an error.

```
EVIDENCE_TIERS.artifact.implemented   // false
evidenceTier('artifact')              // throws
```

Future non-browser option

REA is not included in this build. The browser application produces snapshot and live evidence only.

A future CLI or server deployment could evaluate rea-agents for local artifact inspection.

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Demo and verification

The browser demo uses simulated cluster state. Offline scripts reproduce the rule score and exercise tool execution, fallback and approval boundaries.

live demo

gpu-waste-auditor-2026.web.app

verification scripts

verify_triage.mjs rule baseline and result invariants

verify_pipeline.mjs 81 / 81, scripted transport and in-memory cluster

browser_check.mjs 31 / 31 in the earlier recorded browser check